Advertising Campaign ROI: Identifying High-Profit Customer Targets

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Domain

The focus of this project is the Digital Advertising and Marketing Analytics industry, specifically the economic challenge of optimizing marketing Return on Investment (ROI). In modern business, "wasted" ad spend, money spent on customers who will never convert or will not generate enough revenue to cover the cost of the ad, is a significant drain on profitability. This project moves beyond simple conversion prediction to build a Profitability Classifier that identifies which advertising targets will generate a positive net return, and which will result in a net loss.

Dataset Selection

We will utilize the Bank Marketing Dataset from the UCI Machine Learning Repository, which documents a large-scale direct advertising campaign. The dataset contains 41,188 records and 21 features, including demographic data, previous campaign results, and socioeconomic indices. This dataset contains non-standard missing values (the string "unknown"), categorical variables that require complex encoding, and an "imbalanced" target variable with only 11% of instances successful. Our process involves stripping "unknown" placeholders, performing median imputation, and standardizing features like "age" and "duration" to ensure the ROI model is not biased toward large numerical outliers.

Research Questions

To evaluate the economic efficiency of the advertising campaign, this project seeks to answer the following questions:

1. Which specific advertising channels or customer segments consistently generate a positive net profit versus a net loss?
2. Is there a clear "contact duration" or "economic index" threshold where the probability of generating profit drops to zero?
3. Which customer profiles are "Profit Traps", receiving high engagement and ad spend but generating no net return for the company?

Methodology

Data Engineering & Cleaning

Our methodology begins with a rigorous cleaning pipeline designed to resolve the inherent "messiness" of the raw bank marketing dataset. We first addressed non-standard missing values by replacing all instances of the “unknown” string with null values, followed by median and mode imputation to restore dataset integrity. To ensure the models were not biased by different numerical scales, we applied a StandardScaler to continuous variables such as age and economic indices. Finally, we resolved the significant class imbalance within the dataset by utilizing Random Undersampling to create a balanced training set for more accurate profit targeting.

Feature Engineering

To shift the project's focus from simple conversion to financial ROI, we implemented an extensive feature engineering layer based on specific business logic. We assigned a fixed Acquisition Cost of $5 to every advertising contact and a Conversion Revenue of $100 to successful outcomes to calculate the net financial return. From this calculation, we engineered a new binary target variable, Is_Profitable, which allows the model to distinguish between 'High Profit' and 'Zero Profit' campaigns. We preserved the original campaign outcome ('y') as a secondary target variable to allow for a comparative analysis between traditional conversion success and actual net profitability. Additionally, we utilized One-Hot Encoding to transform categorical variables like job roles and education levels into numerical vectors suitable for mathematical analysis.

Modeling

For the modeling phase, we implemented a comparative approach using three distinct algorithms to identify the most effective predictor of campaign profitability. We utilized Logistic Regression as a baseline to identify linear relationships, while Random Forest was employed to capture more complex, non-linear customer behaviors. Furthermore, we utilized Gradient Boosting to maximize the precision of our predictions and minimize "wasted" ad spend on non-profitable leads. Each model was evaluated based on its F1-Score to ensure a balance between finding profitable opportunities and avoiding costly false positives.

Evaluation

Performance will be assessed using a comprehensive suite of metrics including Accuracy, F1-Score, and AUC-ROC curves to ensure the model effectively distinguishes between profitable and non-profitable segments. We will specifically prioritize the F1-Score for the "Profit" class, as this metric balances the business cost of missing a high-value opportunity against the cost of wasting budget on a non-profitable lead. Additionally, we will generate a Confusion Matrix to visualize the rate of False Positives, ensuring the model provides a reliable strategy for high-precision budget allocation. Finally, Feature Importance plots will be used to validate the model's logic and confirm which demographic drivers are most critical to advertising ROI.

Anticipated Challenges

Class Imbalance

Since most advertising leads do not result in profit, we anticipate the model may struggle with "False Positives." We will mitigate this by undersampling and tuning the probability threshold.

Data Leakage

We must ensure that economic indicators used in the model are "pre-campaign" metrics to ensure the model provides a true prediction rather than a hindsight analysis.

Abstract

This project develops a predictive pipeline to optimize the ROI of advertising campaigns by distinguishing between profitable and non-profitable customer segments. Using a raw dataset of over 41,000 records, the project focuses on cleaning "messy" data and engineering a net profit target variable. By training comparative Random Forest and Gradient Boosting models, we aim to pinpoint the specific demographic and behavioral drivers, such as contact duration and prior campaign success, that drive high-profit outcomes. The final deliverable is an ROI Strategy Map that allows companies to visualize profit probabilities, enabling surgical Precision in budget allocation and significantly reducing wasted marketing expenditure.

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